

Capstone 2 Project Report

Market Reaction to Weather Events

Problem Statement

Weather disasters with exceptionally high property damage are common in the United States. 2024 alone had several events that resulted in damages of \$1 billion or more. I believed these events would have a market reaction. The goal of this project is to predict multiple indices, including market volatility, GDP, unemployment, and the general market, which is measured through the SP500.

This project is the first part in a larger project, the goal of which will be discussed in the results and recommendations portion of this project report. Our goal here is to determine the possible window of time following a meaningful weather event when a reaction is seen in the market indices.

Data Cleaning

Data Sources

1. [FRED database](#) for all economic data. I had to download individual files for each index. They were all in the same format, making cleaning extremely easy. There were some hiccups with establishing a datetime format. Some indices are updated daily, excluding weekends, while others are updated at different frequencies. Overall, it was a clean and convenient data source.
2. The weather data was collected from the [NOAA](#). I removed a majority of the columns. It initially had 28 columns, but ultimately ended up with only 5 columns being used. The location and details of the events weren't necessary. This dataset had multiple entries for some days and none for others. It is aggregated by day in the preprocessing portion of the project. This dataset has mixed dtypes, so quick work was done to make conversion possible, primarily on the property damage column.
3. The tables for each index were merged into a single dataframe, and the weather data was added to it. The NaN values were forward-filled for market data, and weather events were marked as zero for days with no entries.

	State	Event Type	Property Damage	Total Injuries and Deaths
1950-04-28	OKLAHOMA	Tornado	250K	0
1950-04-29	TEXAS	Tornado	25K	0
1950-07-05	PENNSYLVANIA	Tornado	25K	2
1950-07-05	PENNSYLVANIA	Tornado	2.5K	0
1950-07-24	PENNSYLVANIA	Tornado	2.5K	0

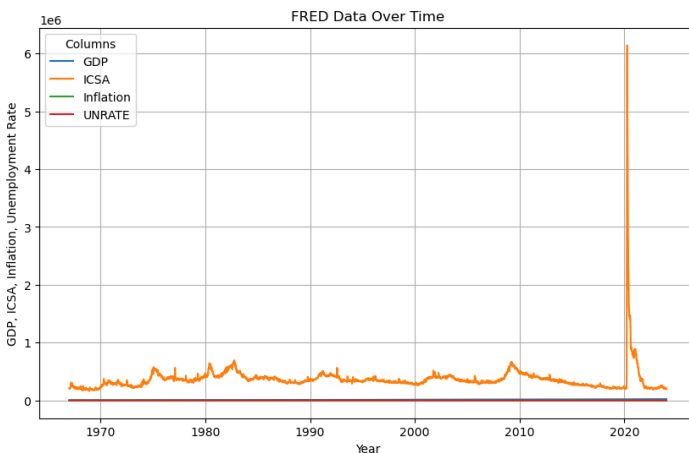
The table above is the original form of the 5 weather columns used in this project. The table below shows the final form of the weather data used. All categorical features were removed. The only categorical feature in this dataset is the 'flagged_event' column, not depicted here.

Property Damage	Total Injuries and Deaths	num_events
3025000.0	7.0	4.0
2500.0	2.0	1.0
500000.0	5.0	2.0
0.0	2.0	1.0
550000.0	24.0	4.0

Exploratory Data Analysis

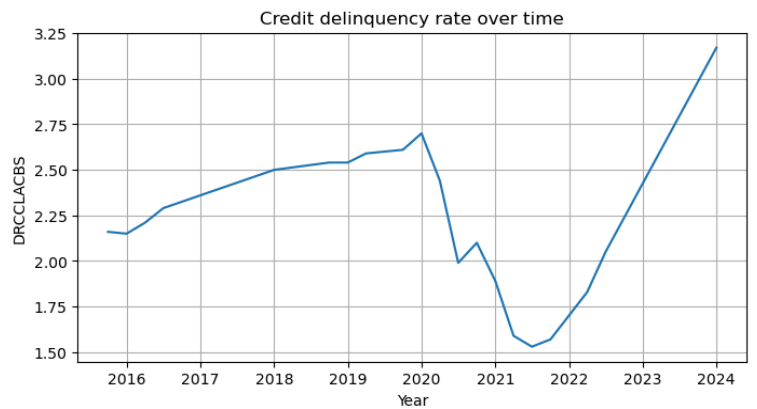
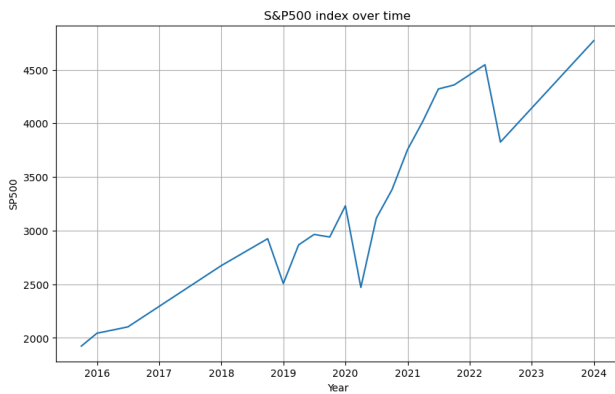
Trends

1. The index with the highest jump is 'ICSA'. It is between the years 2020 and 2021. In fact, many indices show a reaction in this time window. The change in 'ICSA' is on a larger scale than changes in the other indices.

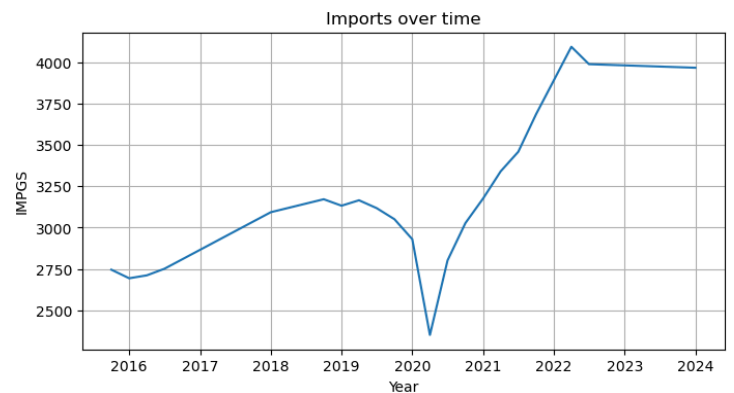
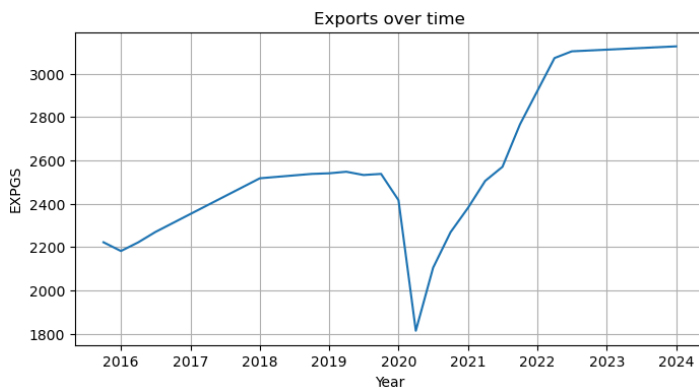


In 2020, we see a very substantial jump in 'ICSA', the same time we see a drop in the 'S&P500'.

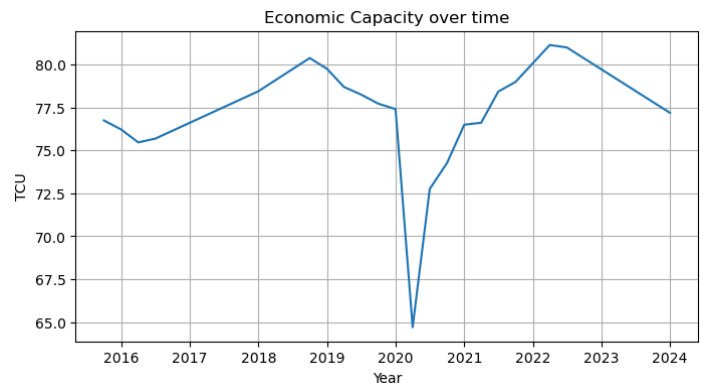
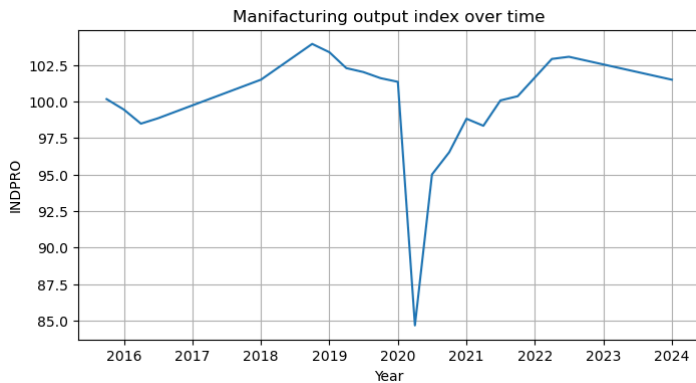
2. From 2021 to 2022, 'S&P500' and credit card delinquency trended in opposite directions. Credit card delinquency rate decreased and the 'S&P500' index remained elevated. This relationship can point to market stability during that point in time. In 2022 we start to see a fall in the index and a rise in credit card delinquency. The 'S&P500' recovers quickly while credit card delinquency continues to rise.



3. Imports and exports have similar trends, which makes sense, but is worth noting for market behavior.



4. Manufacturing remained consistent from 2016 to 2024, with the only large dip occurring when businesses shut down during the initial COVID-19 outbreak (same large reaction seen in the others).



Model Selection and Results

I completed the preprocessing and modeling twice, the first with several models (roughly 26), and the second, I simplified the process by creating a looped grid search for multiple models and targets. That proved to have higher model scoring and made the notebook easier to navigate. A majority of the models explored were regression models. At the end, I captured the window in time where 'flagged_event' (a binary flagged column that indicates if a meaningful weather event has taken place) is most impactful to our highest performing target, 'VIXCLS'.

```
param_grids = {  
    "Random Forest": {"n_estimators": [100, 275], "max_depth": [5, 12]},  
    "XGBoost": {"n_estimators": [100, 275], "learning_rate": [0.01, 0.1]},  
    "Linear Regression": {}  
}
```

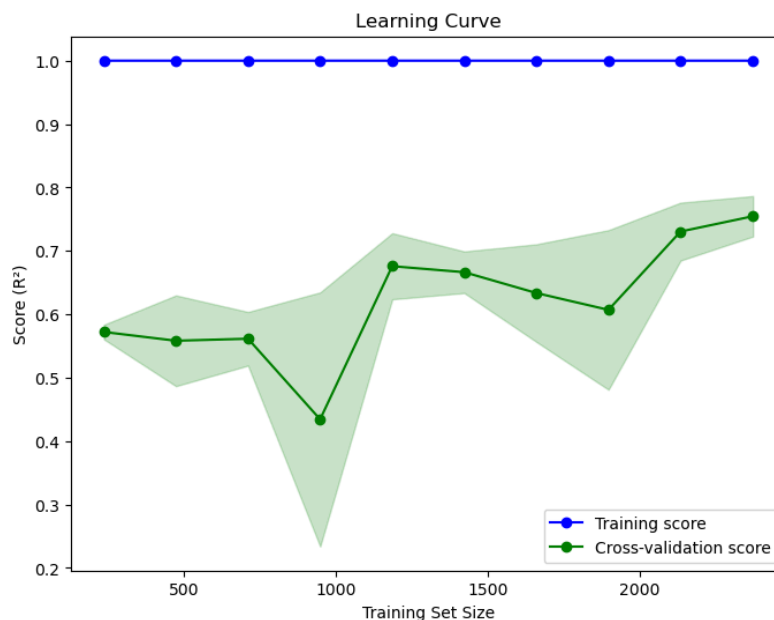
Parameters used for each model type

Group One

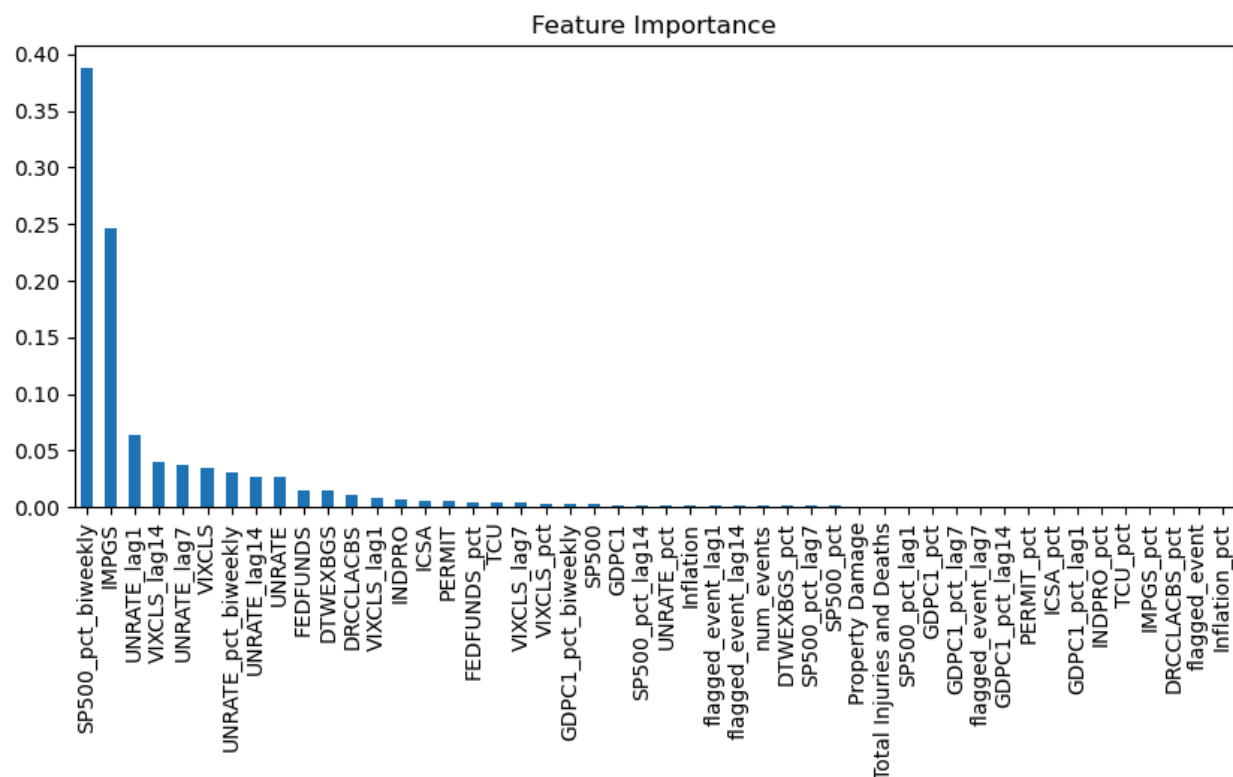
	Target	Model	Best Params	Test R2
4	VIXCLS_pct_biweekly	XGBoost	{'learning_rate': 0.1, 'n_estimators': 275}	0.880441
3	VIXCLS_pct_biweekly	Random Forest	{'max_depth': 12, 'n_estimators': 275}	0.828034
5	VIXCLS_pct_biweekly	Linear Regression	N/A	0.757197
1	SP500_pct_biweekly	XGBoost	{'learning_rate': 0.1, 'n_estimators': 100}	0.645527
0	SP500_pct_biweekly	Random Forest	{'max_depth': 12, 'n_estimators': 275}	0.575503

This is the first round of models with their scores and parameters. 'VIXCLS' was the highest performing target with XGBoost being the best model. The top models were ran individually to check for overfitting. The individual model's score is a bit lower than the looped model found in grid search. This is common throughout the model testing for the rest of the notebook. The window size for this group is 14 days.

The top 2 models use the maximum values for the parameters. It might be beneficial to expand parameters in future modelling, but this runs the risk of overfitting as the model becomes more complex.



This is the graph for model 1. There is some overfitting for sure, but the graph suggests that with more data the model might fit the data better. The blue line is exactly on 1.0 showing that the model is memorizing the training data.

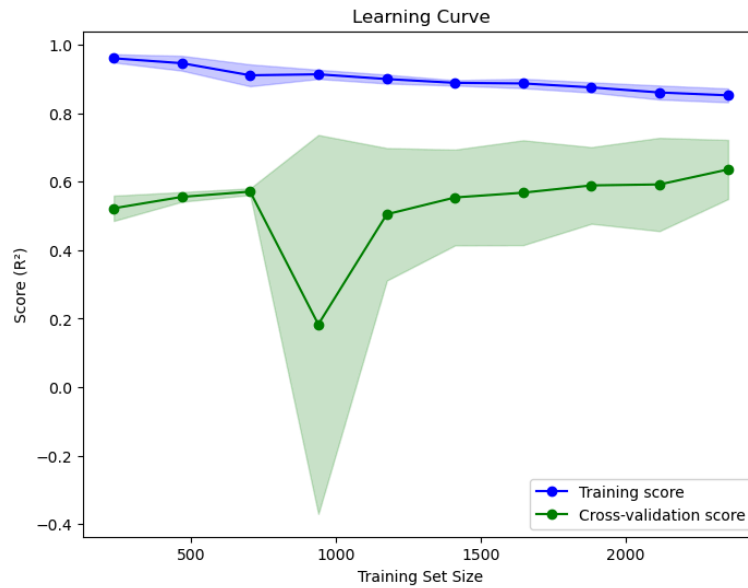


This model's most important features are the biweekly percent change of 'SP500' and imports. The unemployment rate and market volatility are also top features.

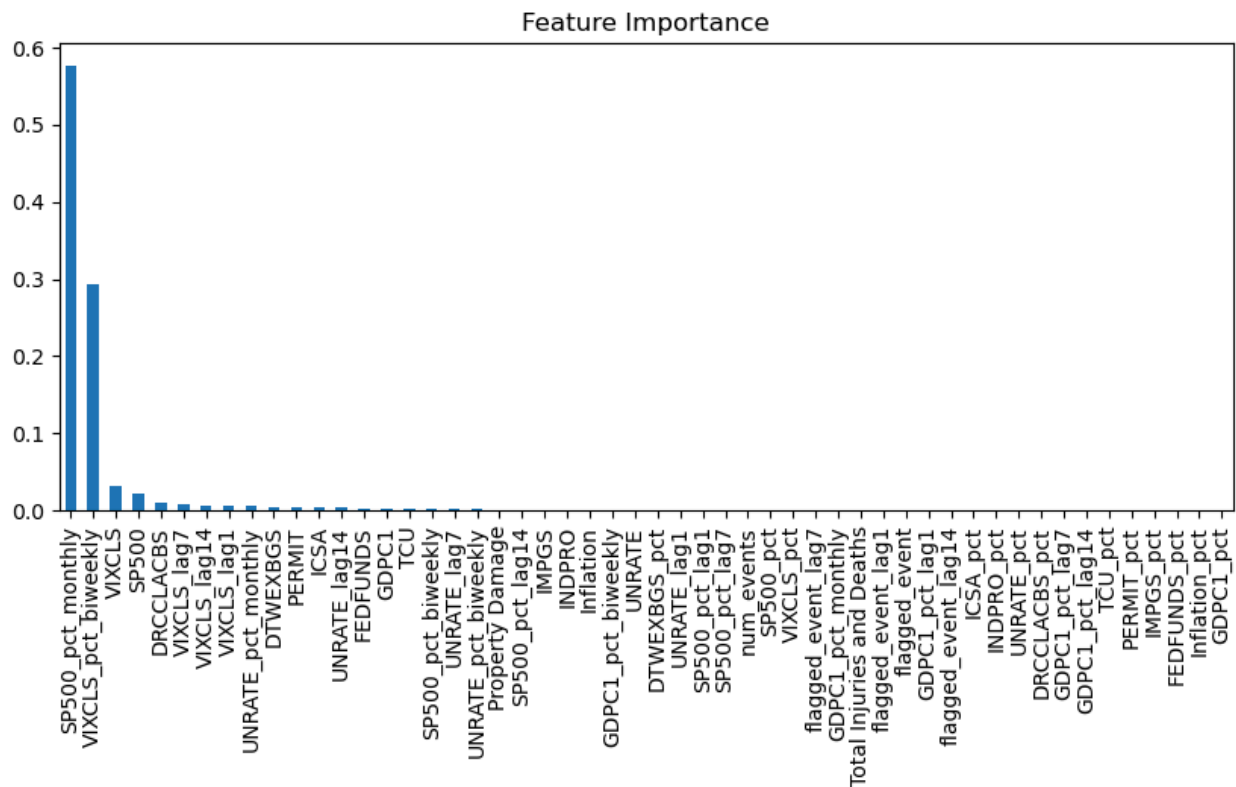
Group Two

	Target	Model	Best Params	Test R2
3	VIXCLS_pct_monthly	Random Forest	{'max_depth': 5, 'n_estimators': 100}	0.671849
1	SP500_pct_monthly	XGBoost	{'learning_rate': 0.1, 'n_estimators': 100}	0.662733
0	SP500_pct_monthly	Random Forest	{'max_depth': 12, 'n_estimators': 100}	0.640979
4	VIXCLS_pct_monthly	XGBoost	{'learning_rate': 0.01, 'n_estimators': 275}	0.636354
2	SP500_pct_monthly	Linear Regression	N/A	0.069402

The window size for group two is 30 days. They performed 20 roughly percentage points lower than the bimonthly group. 'VIXCLS' is still the best target, but 'SP500' takes the 2nd and 3rd top models for this group. There is also more varied parameters instead of the maximum values. There is a steep drop in model performance after model 4.



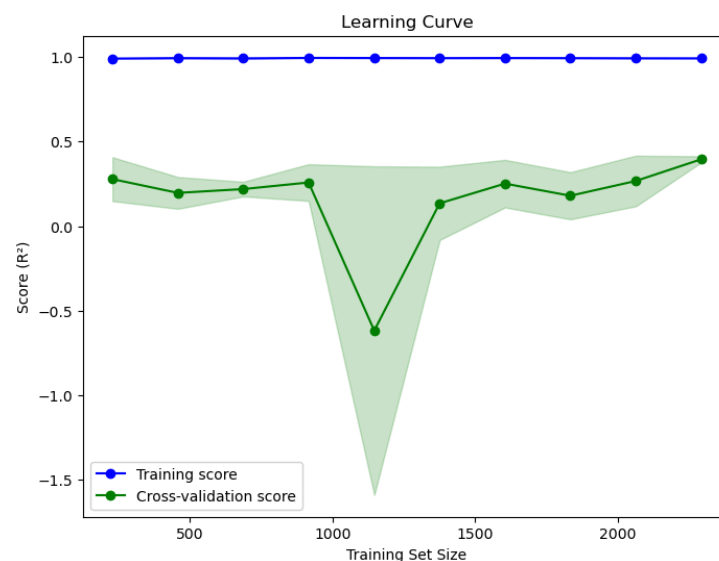
The group 2 model is ranked lower in terms of R^2 score than the group 1 model. Its feature importance is less spread out. It relies mostly on the 'SP500' percent change over the month and the biweekly "VIXCLS" change. The last model also had the percent change of 'SP500' listed as very important.



Group Three

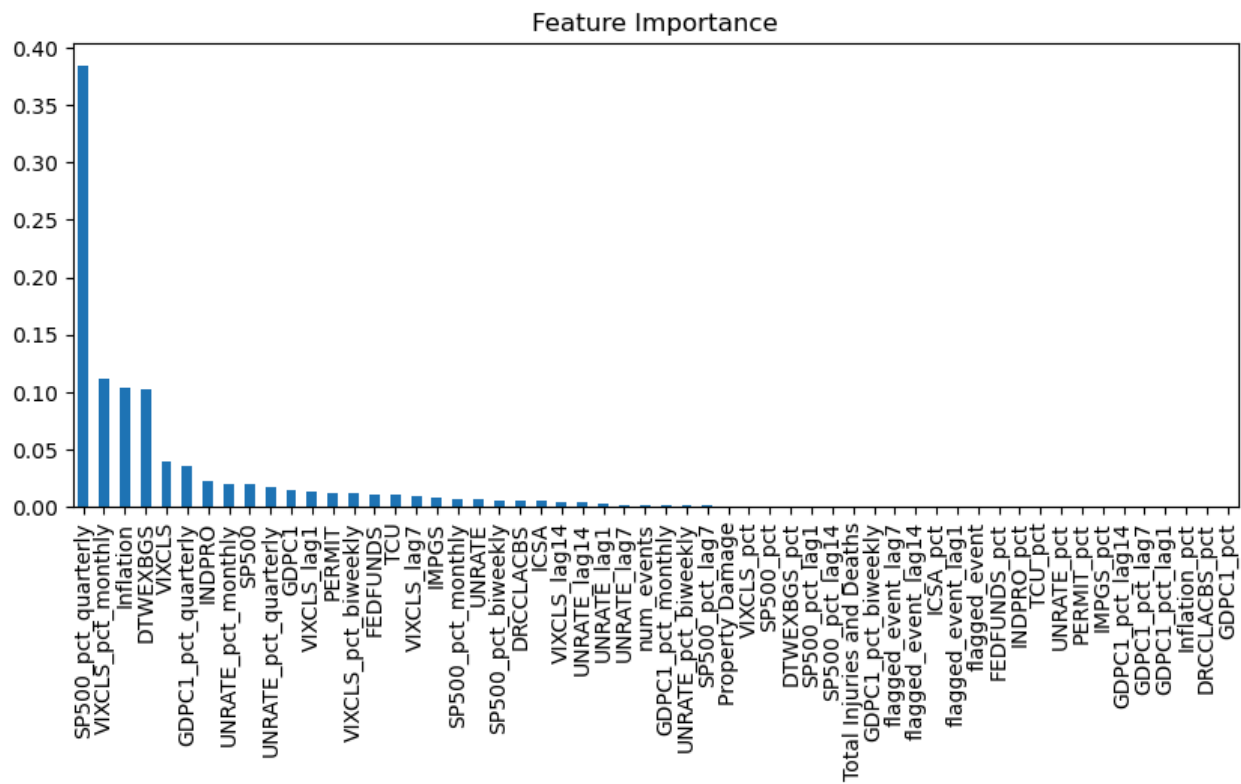
	Target	Model	Best Params	Test R2
5	VIXCLS_pct_quarterly	Linear Regression	N/A	0.684230
1	SP500_pct_quarterly	XGBoost	{'learning_rate': 0.01, 'n_estimators': 100}	0.373055
3	VIXCLS_pct_quarterly	Random Forest	{'max_depth': 12, 'n_estimators': 275}	0.325296
10	UNRATE_pct_quarterly	XGBoost	{'learning_rate': 0.01, 'n_estimators': 275}	0.296645
4	VIXCLS_pct_quarterly	XGBoost	{'learning_rate': 0.01, 'n_estimators': 100}	0.239264

The top model in group three has a decent R2 score, but all models after have scores much lower. Interestingly, Linear Regression is the best model for the quarterly targets. 'VIXCLS' has been the best target for all 3 groups.

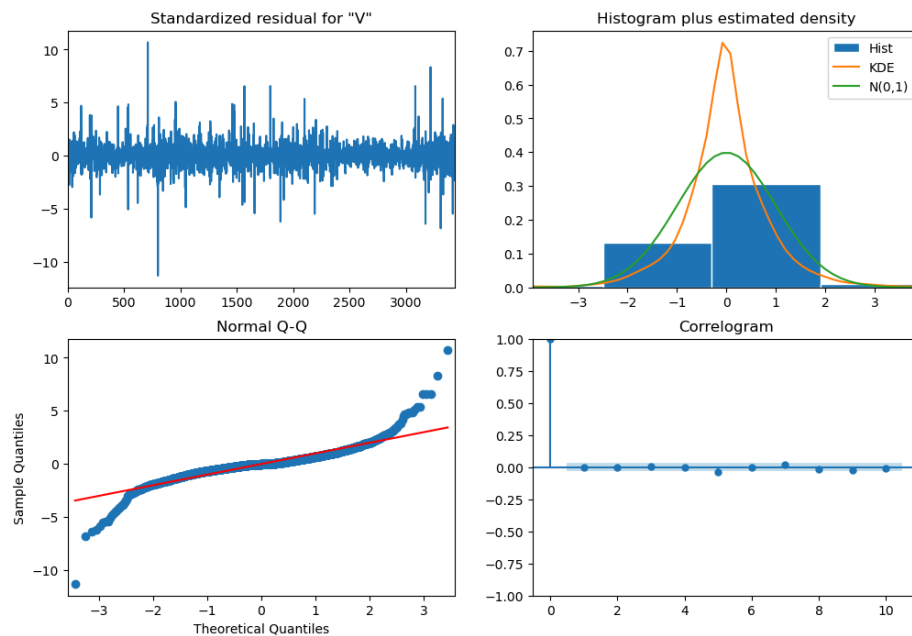


The test score for group 3's top model is very similar to the other 2 models. There is a sharp dip in R2 score one you pass 1,000 samples but the score rebounds as the number of samples increases. Seeing this across all models strongly suggest that more data is needed in future analysis.

At the quarterly level, there is still a strong dependence on features of 'SP500' and 'VIXCLS'. Inflation and credit card delinquency become stronger players in a larger time window. This model has the most evenly distributed feature importance than the previous one.



ARIMA Model



The target for the ARIMA model is 'VIXCLS_pct_quarterly' with the exogenous feature being 'flagged event'. This was to evaluate how 'VIXCLS' changes quarterly in relation to meaningful weather events. The diagnostic graphs above demonstrate model behavior. The correlogram has all points after the first one close to 0 on the y-axis, meaning our residuals are likely uncorrelated. The Normal Q-Q and the histogram show that the residuals are not normally distributed and may underpredict large shifts in 'VIXCLS'.

Recommendations

I would recommend expanding the datasets to strengthen model performance and reduce the risk of overfitting. Across all regression model groups (biweekly, monthly, quarterly), there is a large dip in R2 scores around 1,000 samples, followed by a rebound. This pattern reinforces that the models need more data to generalize effectively. Better generalization leads to the model capturing relationships rather than simply memorizing patterns. This can be achieved using the same datasets since they date back to the 1950s.

Future models can build on relationships identified in this project. It would be worth exploring why certain indices become more important with varying time window sizes. Since 'VIXCLS' and 'SP500' are the best performing targets, it would be useful to use a VAR model in the future to capture the relationship between these two targets over time. Unemployment and GDP were also targets but did not perform well under the parameters tested.

Ongoing data collection and model refinement will help expand this project to improve market forecastability. These models showed sensitivity to a smaller training dataset, so score stability over different sample sizes is recommended to improve outcomes. With these adjustments, the models can be used as tools for reliable market forecasting.